

Business Problem Statement

A global media investment firm wants to optimize its film financing strategy by better understanding the historical factors that drive a movie's financial success and critical acclaim. The management team has observed significant volatility in box office returns across different genres, production companies, and release years. They are particularly interested in uncovering how variables such as production budget, film rating (e.g., R, PG, PG-13), and the involvement of specific directors or stars impact the "Gross" earnings and "Score" of a film.

You are tasked with analyzing the comprehensive movies dataset—spanning four decades of cinematic history—to answer the following overarching business question:

“How can the investment firm leverage historical movie data to identify high-ROI trends, predict audience reception, and optimize future production and distribution portfolios?”

Deliverables

1. **Data Preparation & Modeling (Python):** Clean the raw dataset by handling missing values in the "budget" and "gross" columns, and transform the "released" column into a standardized date format for seasonal analysis.
2. **Data Analysis (SQL):** Organize the movie records into a structured database. Run queries to identify the most profitable genres, determine the correlation between high budgets and high scores, and rank production companies by their historical success rates.
3. **Visualization & Insights:** Build an interactive dashboard that highlights patterns such as the growth of specific genres over time, heatmaps of success by country of origin, and a breakdown of the highest-grossing directors and stars.
4. **Report and Presentation:** Write a clear project report summarizing key findings (e.g., "Action movies released in June have a 20% higher gross on average"). Prepare a presentation that communicates actionable investment recommendations to the firm's stakeholders.